

# Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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## Abstract

Prepending a *primer* of factual sentences about a graph’s structure — each node’s degree, clustering coefficient, random-walk return probability or component — to its text encoding is a natural way to help a language model answer questions about that graph. Across 84,000 generations from four Qwen3 arms (1.7B and 4B, thinking off and on) on six GraphQA tasks and seven edge densities over 40-node Erdős–Rényi graphs, we separate two things a primer can do: determine the answer by itself, which a graph-blind solver detects, and add text, whose cost we fit across five conditions spanning 37 to 4,691 characters. Length turns out to be close to free (a length term of  $-0.3$  points per thousand characters, positive in 8 of 16 cells). Content that determines the answer is not: it is worth about  $+16$  points where the model is failing and nothing where it is not, and on `node_degree` it costs qwen3-4b 12 points at  $p=.50$  while gaining it 27 at  $p=.85$  — same arm, same task, same primer, the sign following the accuracy the cell already had. The generated text shows why: the model stops counting and starts looking up, its median response falling from 273 tokens to 34. Structural side information that does not determine the answer is worth  $+2$  to  $+4$  points in the same window and nothing outside it, and more than half the cells of a standard sweep sit above 0.90 accuracy, where neither can show anything.

## 1 Introduction

Fatemi et al. (2023) show that a language model’s accuracy on graph-reasoning questions varies substantially with the text encoding of the graph. We ask whether prepending a *primer* — a short preamble of derived, node-level facts such as each node’s degree or clustering coefficient — improves the answers, and, when it changes them, what it changes.

Two things have to be separated before that question has an answer. A primer may help because the model reads the answer off it rather than reasoning

over the graph, which is the shortcut-learning concern of Geirhos et al. (2020) and the alternative to reasoning that Dziri et al. (2023) describe. And a primer may move accuracy simply by being there: added context is reported to degrade reasoning on its own (Shi et al., 2023; Levy et al., 2024). We address the first with a graph-blind solver that reads the primer text and never the graph, which says for each condition whether the primer determines the answer by itself; and the second by fitting the effect on primer length across five conditions spanning 37 to 4,691 characters, rather than trusting a single matched control.

Both separations change the conclusion. Length is close to free: fitted within each (arm, task), the length term averages  $-0.3$  points per thousand characters and is positive in 8 of 16 cells. What is not free is content that determines the answer — and the surprise is that it frequently makes the answers worse. On `node_degree` the degree primer prints the requested number verbatim, and qwen3-4b falls from 99 to 87 points at  $p=.50$ , breaking 13 items for each one it fixes. The generated text shows the mechanism: the model stops counting and starts looking up, its median response falling from 273 tokens to 34.

The same primer, the same task and the same arm gain  $+27$  points at  $p=.85$ , where that arm scores 33 without it. What decides the sign is neither the primer nor the task nor the model, but the accuracy the cell already had. Prompt-level graph interventions have been proposed before — Wang et al. (2023) give *Build-a-Graph* and *Algorithmic* prompting, Perozzi et al. (2024) learn a continuous encoding — and what we add is a controlled account of when one helps, when it hurts, and how much of the help is leakage, read from 84,000 generations on 40-node Erdős–Rényi graphs (Erdős and Rényi, 1959).

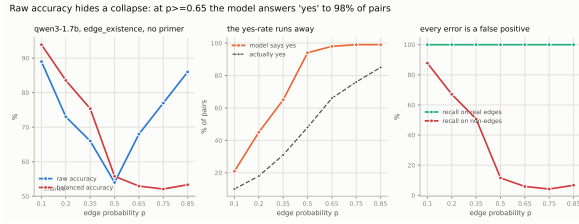


Figure 1: Why edge\_existence needs balanced accuracy. qwen3-1.7b under none: raw accuracy falls to 54% at  $p=.50$  and rises again to 86% at  $p=.85$ , which reads as a recovery. Balanced accuracy over the same responses reaches chance at  $p=.65$  and stays there, because the model answers “yes” to 98–99% of pairs while 66–85% of them are edges.

## 2 Method

### 2.1 Primers and controls

A prompt is a primer, then the graph in the incident encoding, then the question. Seven conditions share one renderer, so they differ in content and never in format: *none*; four content primers reporting node degree, clustering coefficient, random-walk return probability (Dwivedi et al., 2022) and connected-component membership; *all*, which renders degree, clustering and *rwse* together; and *filler*. Mean primer lengths are 37, 891, 1,629, 2,949 and 4,691 characters for components, degree, clustering, *rwse* and *all*, and 1,829 for *filler*.

*filler* is a *degenerate-content* preamble of comparable length to *clustering*: forty sentences of the form “Node  $i$  is simply present within the graph  $G$ ”. It is not content-free — those sentences state the vertex set exactly, and a solver reading only *filler* scores 1.00 on *node\_count* — and it is length-matched to *clustering* alone, so we use it as one condition among several rather than as the reference a content effect is defined against.

The control that does the work is the graph-blind solver. It parses the rendered primer text and applies exact rules, never touching the graph, so its score per (task, condition) is what the primer alone determines. We call a condition *answer-carrying* on a task when that bar is at least 0.85; the bars are bimodal — 18 of the 42 (task, condition) bars sit at exactly 1.000 and the highest of the rest is 0.746 — so any threshold between them gives the same split.

### 2.2 Corpus, arms and measurement

Four arms answer one shared prompt set: Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with native thinking mode off (qwen3-1.7b, qwen3-4b) and on (*-think*). Graphs are Erdős–Rényi with  $n=40$  at  $p \in \{.10, .20, .35, .50\}$ , 100 graphs per (task, density) cell, six GraphQA tasks and seven conditions, extended on *node\_degree* and *edge\_existence* to  $p \in \{.65, .75, .85\}$ . Decoding is greedy. The main sweep ran with an 8,192-token budget on every arm; the high-density extension ran at 2,048 on the plain arms and 8,192 on the thinking arms, so truncation rates are not comparable across those two bands and are reported separately.

Three measurement decisions change what the numbers mean. *Exact match* is the outcome for every task; on *connected\_nodes* the usual set-F1 sits at 0.93–1.00 in every cell while exact match spans 0.48–0.92, so F1 cannot separate the conditions. *Balanced accuracy* accompanies accuracy on *edge\_existence*, where the share of node pairs that are edges rises from 10% to 85% across our density range: raw accuracy recovers with density while balanced accuracy falls to chance (Figure 2). And *truncation is an outcome*, not a missing value, so a pair is dropped when either side truncated — the convention least favourable to the primer — and rates are reported alongside. *node\_count* and *cycle\_check* have a constant gold answer at  $n=40$  and hence a majority-class baseline of 1.000; they are reported in the appendix but excluded from every summary over tasks. Differences are paired on the shared graph, tested with exact McNemar (McNemar, 1947) and corrected with Benjamini–Hochberg within each (arm, task) family. Extraction is not a confound: 99.84% of the 84,000 generations parse, and 99% of the 133 that do not are truncated, 128 of them on *cycle\_check*.

## 3 Results

### 3.1 Most of a standard sweep cannot show an effect

A primer can only move a cell that is not already at floor or ceiling, and most cells are. Of the 324 side-information cells on the four informative tasks, 172 — 53% — sit above 0.90 accuracy without any primer and cannot show a gain by construction. Table 2 bins every cell by that baseline and splits it by whether the primer determines the answer.

| Arm              | Task            | none  | <i>n</i> | $\Delta$ vs none |             |             |              |              |              | $\Delta$ vs filler |              |              |              |              |
|------------------|-----------------|-------|----------|------------------|-------------|-------------|--------------|--------------|--------------|--------------------|--------------|--------------|--------------|--------------|
|                  |                 |       |          | filler           | comp.       | clust.      | rwse         | degree       | all          | comp.              | clust.       | rwse         | degree       | all          |
| qwen3-1.7b       | edge_existence  | 70.5  | 399      | <b>-15.7</b>     | <b>-8.2</b> | +3.8        | +5.3         | +4.0         | <b>+11.5</b> | <b>+7.5</b>        | <b>+19.5</b> | <b>+21.1</b> | <b>+19.8</b> | <b>+27.2</b> |
|                  | node_degree     | 60.5  | 399      | -0.5             | +1.8        | +3.0        | +4.0         | <b>+7.5</b>  | <b>+10.3</b> | +2.3               | +3.5         | +4.8         | <b>+8.0</b>  | <b>+10.8</b> |
|                  | connected_nodes | 72.5  | 399      | <b>-6.2</b>      | +0.8        | +1.3        | +1.0         | +3.5         | +2.0         | <b>+7.0</b>        | <b>+7.5</b>  | <b>+7.3</b>  | <b>+9.8</b>  | <b>+8.3</b>  |
|                  | edge_count      | 1.8   | 211      | -0.4             | -1.4        | -0.4        | -0.8         | +4.0         | +2.4         | -0.4               | -0.4         | -0.4         | <b>+4.5</b>  | +1.3         |
| qwen3-4b         | edge_existence  | 90.5  | 398      | <b>-7.0</b>      | -2.8        | <b>+4.0</b> | <b>-10.0</b> | <b>-4.3</b>  | <b>-9.8</b>  | <b>+4.2</b>        | <b>+11.0</b> | -3.0         | +2.5         | -2.8         |
|                  | node_degree     | 99.2  | 400      | -0.5             | +0.2        | -0.5        | <b>-2.3</b>  | <b>-6.5</b>  | <b>-8.0</b>  | +0.7               | +0.0         | -1.8         | <b>-6.0</b>  | <b>-7.5</b>  |
|                  | connected_nodes | 86.0  | 400      | <b>-12.5</b>     | <b>+9.3</b> | +1.0        | +0.0         | -3.0         | -2.7         | <b>+21.8</b>       | <b>+13.5</b> | <b>+12.5</b> | <b>+9.5</b>  | <b>+9.8</b>  |
|                  | edge_count      | 2.0   | 393      | +1.2             | -1.3        | -1.3        | -1.5         | <b>+27.7</b> | <b>+3.8</b>  | <b>-2.5</b>        | <b>-2.5</b>  | <b>-2.8</b>  | <b>+26.5</b> | +2.5         |
| qwen3-1.7b-think | edge_existence  | 99.2  | 397      | -0.8             | -0.8        | -1.0        | -1.3         | -2.3         | -2.0         | +0.0               | -0.3         | -0.5         | -1.5         | -1.2         |
|                  | node_degree     | 76.8  | 376      | -3.5             | -1.0        | +1.0        | +2.5         | <b>+11.7</b> | <b>+11.6</b> | +2.3               | <b>+4.5</b>  | <b>+5.8</b>  | <b>+14.6</b> | <b>+15.1</b> |
|                  | connected_nodes | 79.7  | 390      | +2.0             | +2.8        | +1.5        | -0.3         | <b>+7.4</b>  | <b>+6.1</b>  | +0.5               | -0.5         | -2.5         | +4.6         | +4.1         |
|                  | edge_count      | 86.3  | 13       | +3.4             | +7.4        | -8.7        | -4.0         | -61.5        | -8.3         | +5.4               | -4.2         | +13.0        | -18.8        | -22.2        |
| qwen3-4b-think   | edge_existence  | 100.0 | 393      | +0.0             | +0.0        | +0.0        | +0.0         | +0.0         | +0.0         | +0.0               | +0.0         | +0.0         | +0.0         | +0.0         |
|                  | node_degree     | 99.7  | 380      | +0.0             | -0.3        | -0.3        | <b>-2.1</b>  | +0.0         | -0.8         | -0.5               | -0.3         | <b>-2.1</b>  | -0.3         | -0.8         |
|                  | connected_nodes | 98.7  | 373      | +0.8             | +0.5        | +0.0        | +0.0         | +0.0         | -0.5         | +0.0               | +0.0         | -0.5         | -0.5         | -0.5         |
|                  | edge_count      | 97.6  | 34       | -2.0             | +1.7        | +1.7        | +0.0         | -23.5        | -41.5        | +2.3               | +0.0         | -6.3         | -19.3        | -36.4        |

Table 1: Main sweep,  $n=40$ , pooled over densities  $p \in \{0.10, 0.20, 0.35, 0.50\}$ , 100 graphs per (task, primer, density) cell before truncation. The none column is exact-match accuracy (%);  $n$  is the smallest number of pairs in the row in which neither side truncated; the rest are paired differences in percentage points, against none on the left and against filler on the right, with truncated generations excluded pairwise. filler is length-matched to clustering only (Section 2.1), so the right half is a length-matched contrast for that column alone. node\_count and cycle\_check are omitted: gold is constant at  $n=40$ , so a shift there is not graph reading (Section 2.2); both appear in Appendix Table 3 onward. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family and baseline,  $q = 0.05$ . A row whose  $n$  is *italic* falls below 100 surviving pairs and is never bolded, because those survivors are selected on whether the generation terminated (Section 2.2).

Both columns show the same shape: nothing at the top of the range, and whatever movement exists concentrated between roughly 0.25 and 0.90.

The two columns differ in size, not in shape. Inside that window a primer that carries the answer is worth about +16 points on average; genuine side information is worth +2 to +4. Above 0.90 both are indistinguishable from zero (-1.4 and -0.8). Per primer, inside the window, only all (+5.3), clustering (+2.9) and rwse (+2.2) are worth anything as side information; components averages -0.3, and its one large effect turns out not to be content at all (Section 3.3).

**Leakage is real but far from complete.** Where the graph-blind solver scores 1.00 — the primer determines the answer outright — the model still has to find it, and mostly does not. On node\_degree the recovered fraction runs from 0.99 on qwen3-4b-think down to 0.43 on qwen3-1.7b; on edge\_count, where the answer requires summing the stated degrees rather than reading one of them, it runs 0.30 on qwen3-4b and 0.006 on qwen3-1.7b under all. A primer that contains the answer is not the same as a model that gives it, and the gap between the two is where the arms differ most.

### 3.2 The same primer helps and harms the same model on the same task

The cleanest demonstration needs no comparison across arms. On node\_degree, qwen3-4b and the degree primer, which prints the queried node’s degree verbatim, the effect tracks the baseline across the seven densities: -6, -1, -7, -12 where the arm scores 99–100 without the primer, then -6 at 82, +7 at 50 and +27 at 33. Same arm, same task, same primer; the sign flips as the task gets hard enough for the model to need it.

The losses are not the cost of the added text. At  $p=.50$  the primer breaks 13 items and fixes 1, and neither cell truncates at all. The response length says what happened: the median falls from 273 new tokens under none to 34 under degree, while filler — longer than degree — leaves it at 266 and costs 2 points. The model abandons a reliable count and reads a number off the primer instead, and the lookup is worse than the count.

### 3.3 What the primer changes in the response

The substitution is visible directly, and it pays when what it displaces was failing. On edge\_count at  $p=.50$ , 7% of qwen3-1.7b’s responses under none state the handshake shortcut — sum the degrees, divide by two — and the rest enumerate edges. Under degree, which lists all 40 degrees,

| Accuracy under                                   | primer carries                       |          | side information |          |
|--------------------------------------------------|--------------------------------------|----------|------------------|----------|
| none                                             | cells                                | $\Delta$ | cells            | $\Delta$ |
| 0.00–0.25                                        | 19                                   | +7.8     | 29               | −1.8     |
| 0.25–0.50                                        | 9                                    | +16.0    | 20               | +2.3     |
| 0.50–0.75                                        | 9                                    | +16.7    | 47               | +3.9     |
| 0.75–0.90                                        | 4                                    | −5.7     | 52               | +0.6     |
| 0.90–1.00                                        | 28                                   | −1.4     | 176              | −0.8     |
| <i>side information only, baseline 0.25–0.90</i> |                                      |          |                  |          |
| components                                       | mean −0.3, median −0.5 over 28 cells |          |                  |          |
| clustering                                       | mean +2.9, median +4.0 over 28 cells |          |                  |          |
| rwse                                             | mean +2.2, median +3.0 over 28 cells |          |                  |          |
| degree                                           | mean +2.0, median +1.5 over 18 cells |          |                  |          |
| all                                              | mean +5.3, median +3.0 over 17 cells |          |                  |          |

Table 2: Mean paired effect against `none` (percentage points) for every (arm, task, density, primer) cell of the main sweep, binned by the cell’s own `none` accuracy and split by whether the primer text alone determines the answer (shortcut bar  $\geq 0.85$ ; Table 13). Constant-gold tasks are excluded. Movement lives in the middle of the baseline range in both columns, and a primer that carries the answer is worth several times one that does not; above 0.90, where most cells sit, neither does anything.

100% state it, and truncation falls from 29% to 20%. At  $p=.10$  the switch runs 67% to 99%. That gain is density-robust rather than a pooled average hiding one cell: +22, +30, +24, +33 across the four main densities.

The other robust gain we measure is not information transfer at all. `components` on `qwen3-4b/connected_nodes` runs +7, +11, +10, +9 from a 37-character preamble — but at  $n=40$  every Erdős–Rényi graph at  $p \geq .20$  is connected, so at three of those four densities the primer emits the same sentence about all 100 graphs, and the gain is larger there than at  $p=.10$ , where 52 of 100 graphs have more than one component and the string actually varies. Whatever that preamble buys, it is not graph-specific content. The same account covers the largest primer effects in the sweep: on `node_count`, whose gold answer is always 40, interposing almost any text between the encoder’s “nodes 0, . . . , 39” preamble and the question interrupts an off-by-one that otherwise costs `qwen3-1.7b` 68.5% of its answers.

The direction of the length change is task-specific, so neither “primers make responses longer” nor its converse holds. On `qwen3-1.7b/edge_existence` the same primers make the model write more (206 tokens under `none` against 352 under `all`); on

`qwen3-4b/connected_nodes` `filler` alone cuts the median from 233 to 94.

Beyond  $p=.50$  on `edge_existence`, `qwen3-1.7b` stops reasoning altogether and answers “yes” to 98–99% of pairs. Neither prompt length nor task difficulty triggers this: a ladder that raises  $n$  at fixed mean degree makes the prompt longer and the task harder without producing it, and on the same graphs the model keeps working on the tasks whose answer is a number rather than a boolean. What the two-valued answer format permits is a cheap guess that is right most of the time — and raw accuracy rewards it while balanced accuracy does not (Figure 2).

### 3.4 Position and composition

**Position inside the primer.** The primer lists nodes in id order, so a retrieval account predicts that accuracy falls as the queried node moves down it. Several primers show steep gradients — `all` on `qwen3-1.7b-think` loses 24.3 points  $[-33.5, -15.2]$  from the first line to the last — and node identity is not the cause, since over 19,600 items a node’s id correlates +0.019 with its degree and +0.020 with its clustering coefficient. But `none` carries no primer and slopes too, by +25.0 points on `qwen3-4b/connected_nodes`, and `filler` has the same forty id-ordered lines as degree, so it is a second retrieval condition rather than a control. Testing each gradient against its own `none` control, only 6 of 48 are distinguishable. Position within the prompt moves accuracy; we cannot attribute it to the primer.

**Bundling.** `all` renders three primers together and is no shorter than their sum, yet across the 41 cells where the parts do anything it delivers a median 0.50 of what they are worth separately  $[0.29, 0.71]$ . A ratio of two noisy quantities is biased downward, so we simulate the matched null — every cell recentred on exact additivity, keeping its own noise scale and the same screen — which gives a median of 1.00  $[0.92, 1.08]$  against the observed 0.50 ( $p < 0.001$ ). On the 19 cells where no single part dominates — the only cells that separate dilution from selection — the ratio to the *sum* is 0.52  $[0.40, 0.71]$  while the ratio to the *largest single part* is 1.00  $[0.74, 1.20]$ . The bundle delivers what its best component delivers.

## 4 Conclusion

Structural primers move GraphQA accuracy mainly by supplying the answer, and only where the model is not already producing it. Where a primer determines the answer it is worth about +16 points inside the difficulty window and nothing above it; genuine structural side information is worth +2 to +4 inside that window and nothing outside it. What the primer does, visible in the generated text, is offer a second procedure that the model takes: on `edge_count` the stated degrees replace enumeration with arithmetic in every response, and on `node_degree` the stated answer replaces a reliable count with an unreliable lookup, costing 12 points on a cell where the model was already right 99 times in 100.

Two measurements make this legible and neither is standard. Fitting the effect on primer length across conditions shows that length costs nothing here, so a single matched control cannot be what a content effect is defined against. And a graph-blind solver, scored per task and condition, separates the gains that are leakage from the gains that are not. Without both, the majority of a standard sweep — more than half of its cells sit above 0.90 — is uninformative about either.

## Limitations

Two model sizes from one family, one encoding, one template, one primer position and greedy decoding throughout. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so the `clustering` primer carries little node-specific information. Most cells hold 100 graphs, which bounds what any single comparison establishes; the claims above are counts and profiles over cells for that reason. The generation budget differs between the main sweep and the high-density extension, so truncation rates are not comparable across them.

Three limits bear on the behavioural results specifically. The procedure a response states is identified by regular expressions over free text, hand-validated on a stratified sample, and a model may use a procedure it does not narrate. Position within the primer is confounded with node identity, and the conditions available as controls carry the same positional structure, which is why we report that effect as unattributed. And the response-length collapse is attributed to the answer format by eliminating prompt length and difficulty, not by manipulating the format directly. Four conditions would

settle the open questions: a shuffled-value primer as a stricter placebo, a corrupted primer to separate copying from computation, single-fact bundles to separate dilution from selection, and a permuted-order primer to break the position/identity confound.

## Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 217,000 generations.

## References

- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2022. Graph neural networks with learnable structural and positional representations. In *International Conference on Learning Representations (ICLR)*.
- Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. 2023. Faith and fate: Limits of transformers on compositionality. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.
- Paul Erdős and Alfréd Rényi. 1959. On random graphs i. *Publicationes Mathematicae Debrecen*, 6:290–297.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2023. Talk like a graph: Encoding graphs for large language models. *arXiv preprint arXiv:2310.04560*.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673.
- Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. Same task, more tokens: the impact of input length on the reasoning performance of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 15339–15353.
- Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.
- Bryan Perozzi, Bahare Fatemi, Dustin Zelle, Anton Tsitsulin, Mehran Kazemi, Rami Al-Rfou, and Jonathan Halcrow. 2024. Let your graph do the talking: Encoding structured data for LLMs. *arXiv preprint arXiv:2402.05862*.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 31210–31227.

Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. Can language models solve graph problems in natural language? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.

## A Supplementary material

### A.1 Example prompt

A clustering-condition node\_degree prompt at  $p=0.10$  (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.

In this graph:

Node 0 is connected to nodes 2, 6, 8, 12, 35.  
Node 1 is connected to nodes 5, 20.

...

Node 39 is connected to nodes 15, 16, 18, 20.

Q: What is the degree of node 16?

A:

### A.2 Per-task results

| Arm    | none | filler       | comp.        | clust.       | rwse        | degree      | all          |
|--------|------|--------------|--------------|--------------|-------------|-------------|--------------|
| 1.7b   | 31.5 | <b>+39.8</b> | <b>+21.2</b> | <b>+67.3</b> | +2.0        | <b>+6.8</b> | <b>+47.0</b> |
| 4b     | 90.2 | <b>+9.3</b>  | <b>+9.0</b>  | <b>+8.8</b>  | <b>+9.3</b> | <b>+8.0</b> | <b>+9.8</b>  |
| 1.7b-t | 98.6 | +1.4         | +0.9         | +1.4         | +1.4        | +0.8        | +1.4         |
| 4b-t   | 99.5 | +0.5         | +0.0         | +0.5         | +0.5        | +0.5        | +0.5         |

Table 3: node\_count, main sweep, pooled over  $p \in \{0.10, 0.20, 0.35, 0.50\}$ . The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. Gold is constant at  $n=40$  on this task, so a shift need not reflect graph reading. **Bold**: exact McNemar, Benjamini–Hochberg within the arm,  $q = 0.05$ .

| Arm    | none  | filler | comp. | clust. | rwse        | degree       | all  |
|--------|-------|--------|-------|--------|-------------|--------------|------|
| 1.7b   | 99.7  | +0.3   | −0.6  | +0.3   | <b>−7.5</b> | <b>−12.6</b> | −1.1 |
| 4b     | 99.7  | −0.3   | −1.3  | +0.3   | +0.3        | −1.5         | −0.5 |
| 1.7b-t | 100.0 | +0.0   | +0.0  | +0.0   | +0.0        | −0.7         | +0.0 |
| 4b-t   | 100.0 | +0.0   | +0.0  | +0.0   | +0.0        | −0.8         | +0.0 |

Table 4: cycle\_check, main sweep, pooled over  $p \in \{0.10, 0.20, 0.35, 0.50\}$ . The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. Gold is constant at  $n=40$  on this task, so a shift need not reflect graph reading. **Bold**: exact McNemar, Benjamini–Hochberg within the arm,  $q = 0.05$ .

| Arm    | none  | filler       | comp.       | clust.      | rwse         | degree      | all          |
|--------|-------|--------------|-------------|-------------|--------------|-------------|--------------|
| 1.7b   | 70.5  | <b>−15.7</b> | <b>−8.2</b> | +3.8        | +5.3         | +4.0        | <b>+11.5</b> |
| 4b     | 90.5  | <b>−7.0</b>  | −2.8        | <b>+4.0</b> | <b>−10.0</b> | <b>−4.3</b> | <b>−9.8</b>  |
| 1.7b-t | 99.2  | −0.8         | −0.8        | −1.0        | −1.3         | −2.3        | −2.0         |
| 4b-t   | 100.0 | +0.0         | +0.0        | +0.0        | +0.0         | +0.0        | +0.0         |

Table 5: edge\_existence, main sweep, pooled over  $p \in \{0.10, 0.20, 0.35, 0.50\}$ . The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm,  $q = 0.05$ .

| Arm    | none | filler | comp. | clust. | rwse        | degree       | all          |
|--------|------|--------|-------|--------|-------------|--------------|--------------|
| 1.7b   | 60.5 | −0.5   | +1.8  | +3.0   | +4.0        | <b>+7.5</b>  | <b>+10.3</b> |
| 4b     | 99.2 | −0.5   | +0.2  | −0.5   | <b>−2.3</b> | <b>−6.5</b>  | <b>−8.0</b>  |
| 1.7b-t | 76.8 | −3.5   | −1.0  | +1.0   | +2.5        | <b>+11.7</b> | <b>+11.6</b> |
| 4b-t   | 99.7 | +0.0   | −0.3  | −0.3   | <b>−2.1</b> | +0.0         | −0.8         |

Table 6: node\_degree, main sweep, pooled over  $p \in \{0.10, 0.20, 0.35, 0.50\}$ . The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm,  $q = 0.05$ .

| Arm    | none | filler       | comp.       | clust. | rwse | degree      | all         |
|--------|------|--------------|-------------|--------|------|-------------|-------------|
| 1.7b   | 72.5 | <b>−6.2</b>  | +0.8        | +1.3   | +1.0 | +3.5        | +2.0        |
| 4b     | 86.0 | <b>−12.5</b> | <b>+9.3</b> | +1.0   | +0.0 | −3.0        | −2.7        |
| 1.7b-t | 79.7 | +2.0         | +2.8        | +1.5   | −0.3 | <b>+7.4</b> | <b>+6.1</b> |
| 4b-t   | 98.7 | +0.8         | +0.5        | +0.0   | +0.0 | +0.0        | −0.5        |

Table 7: connected\_nodes, main sweep, pooled over  $p \in \{0.10, 0.20, 0.35, 0.50\}$ . The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm,  $q = 0.05$ .

| Arm    | none | filler | comp. | clust. | rwse | degree       | all          |
|--------|------|--------|-------|--------|------|--------------|--------------|
| 1.7b   | 1.8  | -0.4   | -1.4  | -0.4   | -0.8 | +4.0         | +2.4         |
| 4b     | 2.0  | +1.2   | -1.3  | -1.3   | -1.5 | <b>+27.7</b> | <b>+3.8</b>  |
| 1.7b-t | 86.3 | +3.4   | +7.4  | -8.7   | -4.0 | <b>-61.5</b> | -8.3         |
| 4b-t   | 97.6 | -2.0   | +1.7  | +1.7   | +0.0 | <b>-23.5</b> | <b>-41.5</b> |

Table 8: edge\_count, main sweep, pooled over  $p \in \{0.10, 0.20, 0.35, 0.50\}$ . The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm,  $q = 0.05$ .

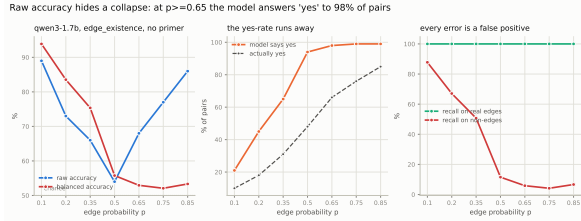


Figure 2: edge\_existence under none, qwen3-1.7b. Accuracy falls to 54% at  $p=.50$  and then rises again to 86% at  $p=.85$ , which reads as a recovery. Balanced accuracy over the same responses does not recover: it reaches chance at  $p=.65$  and stays there, because the model answers “yes” to 98–99% of pairs while 66–85% of them really are edges.

| Arm / task             |               | none | filler | degree | all |
|------------------------|---------------|------|--------|--------|-----|
| 1.7b<br>edge_count     | median tokens | –    | –      | 818    | –   |
|                        | % sum-of-deg. | 7%   | 31%    | 100%   | 40% |
|                        | % truncated   | 29%  | 41%    | 20%    | 44% |
| 4b<br>node_degree      | median tokens | 273  | 266    | 34     | 134 |
|                        | % sum-of-deg. | 0%   | 0%     | 0%     | 0%  |
|                        | % truncated   | 0%   | 0%     | 0%     | 0%  |
| 1.7b<br>edge_existence | median tokens | 206  | 171    | 294    | 352 |
|                        | % sum-of-deg. | 0%   | 0%     | 1%     | 0%  |
|                        | % truncated   | 0%   | 0%     | 0%     | 0%  |
| 4b<br>connected_nodes  | median tokens | 233  | 94     | 258    | 107 |
|                        | % sum-of-deg. | 0%   | 0%     | 0%     | 0%  |
|                        | % truncated   | 0%   | 0%     | 0%     | 0%  |

Table 9: Response behaviour at  $p=0.50$ . Median new tokens is computed over generations that terminated, since a capped response is truncated at the budget rather than long, and is reported as – where more than 20% of generations truncated, because the surviving median is then a sample of the short responses; read the truncation row for those cells. “sum-of-deg.” is the share of responses that state the handshake shortcut (sum the degrees, divide by two) rather than enumerating edges. The primer moves length in both directions depending on the task, and on edge\_count it replaces the method outright.

| Arm                      | Task            | $\sum$ parts | all   | ratio |
|--------------------------|-----------------|--------------|-------|-------|
| <i>other tasks</i>       |                 |              |       |       |
| 1.7b                     | connected_nodes | +5.7         | +2.0  | 0.35  |
| 1.7b                     | edge_existence  | +13.0        | +11.5 | 0.88  |
| 1.7b                     | node_degree     | +14.5        | +10.2 | 0.71  |
| 1.7b-t                   | connected_nodes | +8.6         | +6.1  | 0.71  |
| 1.7b-t                   | edge_existence  | -4.5         | -2.0  | 0.44  |
| 1.7b-t                   | node_degree     | +15.6        | +11.5 | 0.74  |
| 4b                       | edge_existence  | -10.3        | -9.8  | 0.95  |
| 4b                       | node_degree     | -9.2         | -8.0  | 0.86  |
| <i>edge_count</i>        |                 |              |       |       |
| 1.7b-t                   | edge_count      | -84.5        | -22.2 | 0.26  |
| 4b                       | edge_count      | +24.9        | +3.8  | 0.15  |
| 4b-t                     | edge_count      | -22.5        | -41.1 | 1.82  |
| <i>median over cells</i> |                 |              |       | 0.50  |

Table 10: all renders degree, clustering and rwse in one primer, so the parts and the bundle are both measured. The first two columns are paired differences against none in percentage points, averaged over  $p \in \{0.10, 0.20, 0.35, 0.50\}$ ; rows whose parts sum to under 4 points are omitted, since a ratio of two null effects is not informative. The final row is the median over the 41 individual (arm, task, density) cells rather than over the averaged rows above, because averaging numerator and denominator over densities first gives a ratio of means that the largest cells dominate. Its bootstrap interval is  $[0.29, 0.71]$ . We read this as a direction and not as a coefficient: the cells are screened on a noisy denominator, and selecting on a noisy quantity inflates it, so a median of ratios sits below 1 even under exact additivity. The bias runs against the claim, so the bundle falling short of its parts survives it; its size does not (Section 3.4). edge\_count is the widest case: its ratios run 0.26, 0.15, 1.82, from keeping almost none of the effect to overshooting it.

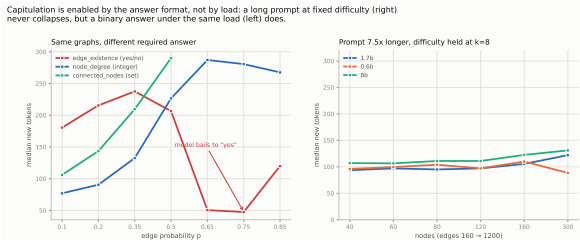


Figure 3: The response-length collapse on edge\_existence and two controls on it. (a) median new tokens by density under none for three tasks answered about the same graphs: only the task with a two-valued answer collapses, while the tasks that must enumerate keep generating, so the trigger is not the graph. (b) median new tokens against  $n$  at fixed mean degree, where the prompt grows and the task gets harder without producing a collapse. What the collapse tracks is the answer format.

| Accuracy without<br>the primer (%) | significant effects |      |      | median<br>$\Delta$ (pts) |
|------------------------------------|---------------------|------|------|--------------------------|
|                                    | cells               | help | harm |                          |
| < 20                               | 12                  | 12   | 0    | +23.0                    |
| 20 – 50                            | 12                  | 11   | 1    | +24.5                    |
| 50 – 80                            | 19                  | 14   | 5    | +18.0                    |
| 80 – 95                            | 7                   | 2    | 5    | –16.0                    |
| > 95                               | 2                   | 0    | 2    | –13.5                    |

Table 11: Every significant primer effect on the four non-constant-gold tasks, across all four arms and all densities, grouped by the accuracy that (arm, task, density) cell reached *without* a primer. Effects are paired differences against `none` in percentage points, exact McNemar with Benjamini–Hochberg correction, cells of at least 100 pairs. Below 80% a primer almost always helps and the median gain is around 20 points; above it a primer usually hurts, by a similar margin. Scoring truncated generations as wrong rather than dropping them changes the counts but not the crossover.

### A.3 Response behaviour

### A.4 Where effects live

### A.5 Robustness analyses

| Arm              | node_count | node_degree | edge_count | connected_nodes | cycle_check | all   | cap    |
|------------------|------------|-------------|------------|-----------------|-------------|-------|--------|
| gemma4-12b       | 1.00       | 1.00        | 0.93       | 1.00            | 1.00        | 0.987 | 0/150  |
| gemma4-12b-think | 1.00       | 1.00        | 1.00       | 1.00            | 1.00        | 1.000 | 43/150 |
| gemma4-e4b       | 1.00       | 1.00        | 0.96       | 1.00            | 0.97        | 0.986 | 5/150  |
| gemma4-e4b-think | 0.97       | 1.00        | 0.93       | 1.00            | 0.97        | 0.973 | 0/150  |
| qwen3-8b         | 1.00       | 1.00        | 0.43       | 1.00            | 0.97        | 0.880 | 0/150  |
| qwen3-8b-think   | 1.00       | 1.00        | 1.00       | 1.00            | 1.00        | 1.000 | 7/150  |
| qwen3-14b        | 0.97       | 1.00        | 0.40       | 1.00            | 0.97        | 0.867 | 0/150  |
| qwen3-14b-think  | 1.00       | 1.00        | 1.00       | 0.97            | 1.00        | 0.993 | 2/150  |

Table 15: The published GraphQA split (5–19 nodes, 30 graphs per cell) under `none`, exact match. Every arm scores 0.97–1.00 on four of the five tasks; only `edge_count` separates them, and only for the two plain Qwen arms (0.40–0.43 against 0.93–1.00 for the rest). `cap` counts generations that hit the token budget, excluded from the accuracies. This corpus cannot measure a primer effect, which is why the main sweep moves to  $n=40$ .

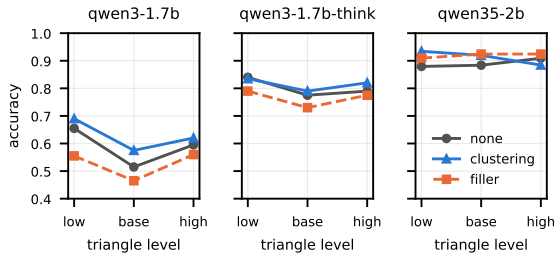


Figure 5: Rewiring: `node_degree` accuracy by triangle level (200 graphs per level; degree sequence and prompt length fixed; mean clustering 0.01, 0.31, 0.73).

### A.6 Per-cell results

Paired effect sizes for every (arm, task, primer) cell:  $\Delta$  in percentage points against `none`, an ex-

| Primer / task                                          | 1.7b  | 4b    | 1.7b-t | 4b-t  |
|--------------------------------------------------------|-------|-------|--------|-------|
| <i>matched: the primer carries what the task needs</i> |       |       |        |       |
| clustering / edge_existence                            | +19.5 | +11.0 | –0.2   | +0.0  |
| components / connected_nodes                           | +7.0  | +21.7 | +0.5   | –0.0  |
| degree / edge_count                                    | +4.9  | +26.4 | –18.8  | –14.5 |
| degree / node_degree                                   | +8.0  | –6.0  | +15.0  | –0.2  |
| <i>mismatched, mean over all pairs</i>                 |       |       |        |       |
| —                                                      | +8.2  | +2.0  | +2.3   | –2.4  |

Table 12: Each primer reports one graph feature and each task needs one, so a pair is *matched* when the primer carries what the task asks for. Entries are paired differences against `filler` in percentage points, averaged over  $p \in \{0.10, 0.20, 0.35, 0.50\}$ ; constant-gold tasks are excluded. The three largest gains we measure are all matched pairs, but matching is not sufficient and not necessary: mismatched content is still worth +8.2 points on qwen3-1.7b, more than two of the matched pairs, and the fourth matched pair reverses with model size. That pair is `degree` on `node_degree`, the one case where the primer states the requested number outright: it gains 8 points on the 1.7B arm and loses 6 on the 4B arm (Section 3.4).

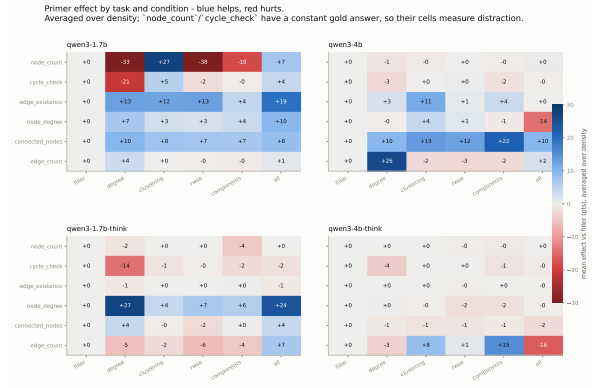


Figure 4: Mean paired effect against `filler` by task and primer, one panel per arm, on a shared symmetric scale so that zero is the same colour throughout. This covers all six tasks and all seven densities and scores truncated generations incorrect, so it is broader than Table 1 and its cells do not match it. The point of the figure is that the sign is not a property of a primer: rows and columns both change colour.

| Task            | none | comp. | clust. | rwse | degree      | filler | all         |
|-----------------|------|-------|--------|------|-------------|--------|-------------|
| node_count      | 1.00 | 1.00  | 1.00   | 1.00 | 1.00        | 1.00   | 1.00        |
| edge_count      | .030 | .030  | .104   | .030 | <b>1.00</b> | .030   | <b>1.00</b> |
| node_degree     | .144 | .144  | .144   | .210 | <b>1.00</b> | .144   | <b>1.00</b> |
| connected_nodes | .000 | .000  | .000   | .001 | .001        | .000   | .008        |
| edge_existence  | .735 | .735  | .727   | .554 | .746        | .735   | .741        |
| cycle_check     | 1.00 | 1.00  | 1.00   | 1.00 | 1.00        | 1.00   | 1.00        |

Table 13: Shortcut bars at  $n=40$  (mean over seven densities): accuracy of a solver that reads only the rendered primer, never the graph. **Bold**: the primer content alone determines the answer. The unbolded 1.00 entries reflect a constant gold answer, not primer content.

| $n$    | $\bar{d}$ | qwen3-1.7b |        |          | qwen3-8b            |          |
|--------|-----------|------------|--------|----------|---------------------|----------|
|        |           | none       | clust. | $\Delta$ | none                | $\Delta$ |
| 20     | 8         | 0.912      | 0.865  | -4.8*    | 0.998               | +0.2     |
| 40     | 8         | 0.728      | 0.770  | +4.2     | 0.995               | +0.0     |
| 80     | 8         | 0.645      | 0.750  | +10.5*   | 0.965               | -0.8     |
| 160    | 8         | 0.624      | 0.693  | +6.9*    | 0.897               | -2.8     |
| 20     | 16        | 0.575      | 0.740  | +16.5*   | 0.985               | -0.2     |
| 40     | 16        | 0.398      | 0.476  | +7.8*    | 0.975               | +1.0     |
| 80     | 16        | 0.225      | 0.323  | +9.8*    | 0.895               | -1.8     |
| 160    | 16        | 0.224      | 0.211  | -1.3     | 0.802               | -4.0     |
| pooled |           | 0.542      | 0.604  | +6.2     | $n=3191, p<10^{-4}$ |          |

Table 14: Fixed-mean-degree sweep on node\_degree: mean degree  $\bar{d}$  is held while  $n$  varies, so density falls as the prompt lengthens. Growing  $n$  eightfold at  $\bar{d}\approx 8$  costs 29 points and saturates after  $n=80$ ; doubling  $\bar{d}$  costs 33–42 points at every  $n$ .  $\Delta$  is clustering against none, exact match. \*: Benjamini–Hochberg at  $q=0.05$  across the eight cells. This corpus has no filler arm, so  $\Delta$  is a net effect. qwen3-8b is a ceiling control.

| Task            | none | comp.       | clust.      | rwse        | degree      | filler      | all         |
|-----------------|------|-------------|-------------|-------------|-------------|-------------|-------------|
| node_count      | .064 | .076        | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> |
| edge_count      | .018 | .018        | .148        | .018        | <b>1.00</b> | .018        | <b>1.00</b> |
| node_degree     | .082 | .082        | .082        | .616        | <b>1.00</b> | .082        | <b>1.00</b> |
| connected_nodes | .082 | .082        | .082        | .082        | .208        | .082        | .352        |
| edge_existence  | .498 | .498        | .720        | .642        | .794        | .498        | .794        |
| cycle_check     | .832 | <b>1.00</b> | .832        | .832        | .946        | .832        | .946        |

Table 16: Shortcut bars on 500 graphs from the published 5–19-node generator, used for the screened arms. At  $n=40$  (Table 13) the edge\_existence/none bar rises to 0.735 through class imbalance, and degree’s gap over none on that task falls from +0.296 to +0.011.

act 95% interval conditional on the discordant pairs (Clopper–Pearson, inverting the same exact McNemar test used for the per-cell  $p$ -values), the permutation  $p$ -value, and  $\Delta$  under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 2.2).  $n$  is the number of pairs in which neither arm truncated. “n/a” marks a cell in which no pair disagreed: the conditional estimate is then exactly zero with no width, which is a statement about this sample and not a bound on the effect, so the discordant count is reported instead.

| Task            | 1.7B  | 1.7B-th.     | 4B   | 4B-th.       |
|-----------------|-------|--------------|------|--------------|
| edge_count      | 22.4% | <b>81.0%</b> | 0.6% | <b>62.2%</b> |
| cycle_check     | 6.1%  | 6.5%         | 1.0% | 1.9%         |
| node_degree     | 0.0%  | 1.1%         | 0.0% | 1.6%         |
| connected_nodes | 0.0%  | 0.8%         | 0.0% | 3.5%         |
| edge_existence  | 0.0%  | 0.2%         | 0.1% | 0.8%         |
| node_count      | 0.0%  | 2.5%         | 0.0% | 0.1%         |

Table 17: Truncation rate (share of main-sweep generations reaching the 8,192-token output budget), by task and arm, pooled over condition and density. **Bold:** above 50%.

| Arm        | Cond.      | hit rate | false alarm |
|------------|------------|----------|-------------|
| qwen3-1.7b | none       | 100.0%   | 40.3%       |
|            | filler     | 100.0%   | 61.8%       |
|            | clustering | 100.0%   | 35.2%       |
| qwen3-4b   | none       | 99.1%    | 12.6%       |
|            | filler     | 100.0%   | 22.5%       |
|            | clustering | 100.0%   | 7.5%        |

Table 18: edge\_existence, pooled over the four main-sweep densities: hit rate (correct on gold “yes”) and false-alarm rate (“yes” on gold “no”). Accuracy changes almost entirely through the false-alarm rate.

| Cond.      | qwen3-1.7b  |                |            |       | qwen3-4b       |       |            |  |
|------------|-------------|----------------|------------|-------|----------------|-------|------------|--|
|            | $\Delta d'$ |                | $\Delta c$ |       | $\Delta d'$    |       | $\Delta c$ |  |
| components | -0.36       | [-0.56, -0.21] | -0.18      | +0.07 | [-0.24, +0.33] | -0.10 |            |  |
| clustering | +0.21       | [+0.03, +0.41] | +0.11      | +0.43 | [+0.12, +0.75] | +0.08 |            |  |
| rwse       | +0.07       | [-0.19, +0.43] | +0.28      | -0.38 | [-0.68, -0.15] | -0.32 |            |  |
| degree     | +0.15       | [-0.05, +0.41] | +0.19      | -0.04 | [-0.33, +0.21] | -0.15 |            |  |
| filler     | -0.93       | [-1.16, -0.71] | -0.46      | -0.25 | [-0.54, +0.00] | -0.25 |            |  |
| all        | +0.55       | [+0.35, +0.77] | +0.27      | -0.36 | [-0.71, -0.11] | -0.31 |            |  |

Table 19: edge\_existence, main sweep: change in sensitivity  $d'$  and criterion  $c$  against none, averaged over the four densities (log-linear correction; 95% bootstrap CI for  $\Delta d'$ ). With the hit rate at ceiling,  $d'$  and  $c$  are not separately identifiable.

| Comparison                 | $\Delta$ | $n$  | density | $p$                 |
|----------------------------|----------|------|---------|---------------------|
| clust. vs. none (main)     | +3.0     | 400  | .10–.50 | 0.26                |
| clust. vs. none            | +1.9     | 2800 | .10–.85 | 0.022               |
| filler vs. none            | -2.4     | 2800 | .10–.85 | 0.0013              |
| clust. vs. filler          | +4.3     | 2800 | .10–.85 | $2.4\times 10^{-8}$ |
| clust. vs. none            | +7.5     | 400  | .35     | 0.0089              |
| clust. vs. none (new seed) | +4.2     | 1600 | .10–.50 | 0.00055             |

Table 20: qwen3-1.7b, node\_degree. All rows except the first come from the dedicated 400-graph-per-density sweep; the last row uses newly seeded graphs. Per-density values are in Appendix Table 21.

| $p$ | clust.—none | filler—none | clust.—filler |
|-----|-------------|-------------|---------------|
| .10 | +0.3        | +3.0*       | −2.8*         |
| .20 | +3.3        | +0.5        | +2.8          |
| .35 | +7.5*       | +1.0        | +6.5*         |
| .50 | +4.3        | −2.0        | +6.3*         |
| .65 | −2.0        | −9.5*       | +7.5*         |
| .75 | +0.5        | −6.3*       | +6.8*         |
| .85 | −0.5        | −3.5*       | +3.0*         |

Table 21: qwen3-1.7b, node\_degree, dedicated sweep, 400 graphs per density. \*: McNemar  $p < 0.05$  (uncorrected).

| Arm              | low   | base  | high  |
|------------------|-------|-------|-------|
| predicted shift  | −11.5 | 0.0   | +16.7 |
| qwen3-1.7b       | +0.03 | +0.28 | −0.02 |
| qwen3-1.7b-think | +0.01 | +0.04 | +0.01 |
| qwen35-2b        | −0.29 | −0.26 | −0.16 |

Table 22: Density-prior test on the rewiring corpus: mean paired shift in signed node\_degree error, clustering minus none (errors clipped at  $\pm 20$ , 200 graphs per level). “Predicted” is the shift if answers moved fully to  $\bar{C}(n-1)$ .

| Task            | 1.7B, acc. under none (%) |       |            | 4B, acc. under none (%) |       |            |
|-----------------|---------------------------|-------|------------|-------------------------|-------|------------|
|                 | plain                     | think | think@2048 | plain                   | think | think@2048 |
| connected_nodes | 72.5                      | 78.7  | 67.2       | 86.0                    | 95.5  | 89.0       |
| cycle_check     | 90.7                      | 97.0  | 38.0       | 99.0                    | 99.3  | 62.0       |
| edge_count      | 1.3                       | 15.8  | 0.0        | 2.0                     | 20.0  | 0.0        |
| edge_existence  | 70.5                      | 99.0  | 97.8       | 90.5                    | 99.8  | 99.8       |
| node_count      | 31.5                      | 91.0  | 86.0       | 90.2                    | 99.5  | 94.8       |
| node_degree     | 60.5                      | 76.2  | 67.7       | 99.3                    | 97.0  | 91.2       |

Table 23: Budget-matched comparison. Every arm ran at 8,192 tokens; “think” is that run and “think@2048” rescores it at 2,048, which is exact under greedy decoding. The plain columns are the 8,192 run; rescoreing them at 2,048 moves them by at most 0.2 points, because the plain arms exceed that budget on almost no terminating generation, so the comparison is matched in effect. Truncated generations are scored incorrect here, so values differ from the dropped-pair convention of Table 1.

| Arm  | Task        | none  | comp. | clust. | rwse | degree | filler | all  |
|------|-------------|-------|-------|--------|------|--------|--------|------|
| 1.7B | node_count  | 0.69  | 0.47  | 0.01   | 0.67 | 0.62   | 0.29   | 0.22 |
|      | edge_count  | 133.4 | 98.1  | 105.7  | 79.0 | 47.7   | 181.2  | 94.2 |
|      | node_degree | 0.94  | 0.91  | 0.78   | 0.91 | 0.67   | 1.27   | 0.91 |
| 4B   | node_count  | 0.10  | 0.01  | 0.01   | 0.01 | 0.02   | 0.01   | 0.00 |
|      | edge_count  | 39.2  | 40.1  | 43.4   | 56.0 | 40.4   | 29.3   | 28.1 |
|      | node_degree | 0.02  | 0.01  | 0.02   | 0.05 | 0.14   | 0.02   | 0.49 |

Table 24: Mean absolute error on the integer tasks, plain arms, main sweep (non-truncated rows).

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 400 | +0.8     | [−3.1, +4.5]   | 0.7861   | +0.8   |
|                 | clustering | 400 | +1.2     | [−3.3, +5.7]   | 0.6458   | +1.2   |
|                 | rwse       | 400 | +1.0     | [−3.5, +5.5]   | 0.7304   | +1.0   |
|                 | degree     | 400 | +3.5     | [−0.8, +7.5]   | 0.1239   | +3.5   |
|                 | filler     | 400 | −6.2     | [−10.0, −1.9]  | 0.0040   | −6.2   |
|                 | all        | 399 | +2.0     | [−2.9, +6.8]   | 0.4644   | +2.0   |
| cycle_check     | components | 343 | −0.6     | [−0.6, +0.4]   | 0.5033   | +2.0   |
|                 | clustering | 362 | +0.3     | [−0.3, +0.3]   | 1.0000   | +8.5   |
|                 | rwse       | 358 | −7.5     | [−8.1, −5.2]   | 0.0001   | −0.2   |
|                 | degree     | 301 | −12.6    | [−12.6, −10.3] | 0.0001   | −17.5  |
|                 | filler     | 341 | +0.3     | [−0.3, +0.3]   | 1.0000   | +3.0   |
|                 | all        | 362 | −1.1     | [−1.6, +0.5]   | 0.2210   | +7.0   |
| edge_count      | components | 211 | −1.4     | [−2.3, +1.0]   | 0.3783   | −1.0   |
|                 | clustering | 238 | −0.4     | [−2.4, +1.9]   | 1.0000   | −0.5   |
|                 | rwse       | 239 | −0.8     | [−2.3, +1.4]   | 0.6837   | −0.8   |
|                 | degree     | 253 | +4.0     | [+0.6, +5.8]   | 0.0179   | +3.0   |
|                 | filler     | 224 | −0.4     | [−2.0, +1.6]   | 1.0000   | −0.5   |
|                 | all        | 211 | +2.4     | [−1.1, +4.6]   | 0.2269   | +0.8   |
| edge_existence  | components | 400 | −8.2     | [−10.9, −4.7]  | 0.0001   | −8.2   |
|                 | clustering | 400 | +3.8     | [−0.9, +8.1]   | 0.1199   | +3.8   |
|                 | rwse       | 399 | +5.3     | [+0.4, +9.8]   | 0.0321   | +5.0   |
|                 | degree     | 400 | +4.0     | [−0.7, +8.4]   | 0.0957   | +4.0   |
|                 | filler     | 400 | −15.8    | [−18.3, −12.0] | 0.0001   | −15.8  |
|                 | all        | 400 | +11.5    | [+6.5, +15.8]  | 0.0001   | +11.5  |
| node_count      | components | 400 | +21.2    | [+15.9, +25.6] | 0.0001   | +21.2  |
|                 | clustering | 400 | +67.2    | [+65.4, +67.2] | 0.0001   | +67.2  |
|                 | rwse       | 400 | +2.0     | [−4.5, +8.5]   | 0.5871   | +2.0   |
|                 | degree     | 400 | +6.8     | [+2.9, +9.9]   | 0.0008   | +6.8   |
|                 | filler     | 400 | +39.8    | [+37.2, +40.6] | 0.0001   | +39.8  |
|                 | all        | 400 | +47.0    | [+43.8, +48.7] | 0.0001   | +47.0  |
| node_degree     | components | 400 | +1.8     | [−2.4, +5.8]   | 0.4599   | +1.8   |
|                 | clustering | 400 | +3.0     | [−2.0, +7.8]   | 0.2577   | +3.0   |
|                 | rwse       | 399 | +4.0     | [−0.9, +8.6]   | 0.1104   | +4.0   |
|                 | degree     | 400 | +7.5     | [+2.8, +11.6]  | 0.0018   | +7.5   |
|                 | filler     | 400 | −0.5     | [−4.6, +3.7]   | 0.9028   | −0.5   |
|                 | all        | 400 | +10.2    | [+4.9, +15.0]  | 0.0003   | +10.2  |

Table 25: qwen3-1.7b: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 392 | +2.8     | [−0.7, +5.9]   | 0.1268   | +2.5   |
|                 | clustering | 394 | +1.5     | [−2.7, +5.6]   | 0.5384   | +2.0   |
|                 | rwse       | 393 | −0.3     | [−4.3, +3.8]   | 1.0000   | +0.2   |
|                 | degree     | 390 | +7.4     | [+3.5, +10.6]  | 0.0001   | +6.2   |
|                 | filler     | 394 | +2.0     | [−1.7, +5.5]   | 0.3275   | +2.5   |
|                 | all        | 392 | +6.1     | [+2.2, +9.4]   | 0.0027   | +6.2   |
| cycle_check     | components | 346 | +0.0     | n/a (0 disc.)  | 1.0000   | −7.5   |
|                 | clustering | 376 | +0.0     | n/a (0 disc.)  | 1.0000   | −0.5   |
|                 | rwse       | 386 | +0.0     | n/a (0 disc.)  | 1.0000   | +2.2   |
|                 | degree     | 295 | −0.7     | [−0.7, +0.5]   | 0.5036   | −21.8  |
|                 | filler     | 388 | +0.0     | n/a (0 disc.)  | 1.0000   | +3.0   |
|                 | all        | 373 | +0.0     | n/a (0 disc.)  | 1.0000   | −0.8   |
| edge_count      | components | 27  | +7.4     | [−5.1, +7.4]   | 0.5037   | −0.8   |
|                 | clustering | 23  | −8.7     | [−17.2, +10.6] | 0.6251   | −1.2   |
|                 | rwse       | 25  | −4.0     | [−4.0, +3.8]   | 1.0000   | −3.2   |
|                 | degree     | 13  | −61.5    | [−61.5, −16.1] | 0.0102   | −3.2   |
|                 | filler     | 29  | +3.4     | [−8.4, +10.2]  | 1.0000   | −1.5   |
|                 | all        | 24  | −8.3     | [−22.8, +13.9] | 0.6794   | +5.0   |
| edge_existence  | components | 398 | −0.8     | [−1.9, +0.9]   | 0.5090   | −0.8   |
|                 | clustering | 398 | −1.0     | [−1.9, +0.6]   | 0.2911   | −1.0   |
|                 | rwse       | 398 | −1.3     | [−2.4, +0.6]   | 0.2274   | −1.2   |
|                 | degree     | 397 | −2.3     | [−3.1, −0.3]   | 0.0249   | −2.5   |
|                 | filler     | 399 | −0.8     | [−1.9, +0.9]   | 0.5051   | −0.5   |
|                 | all        | 399 | −2.0     | [−2.9, −0.1]   | 0.0374   | −1.8   |
| node_count      | components | 342 | +0.9     | [−0.6, +1.4]   | 0.3758   | +1.5   |
|                 | clustering | 369 | +1.4     | [−0.1, +1.4]   | 0.0597   | +9.0   |
|                 | rwse       | 369 | +1.4     | [−0.1, +1.4]   | 0.0597   | +9.0   |
|                 | degree     | 360 | +0.8     | [−0.8, +1.8]   | 0.4512   | +5.5   |
|                 | filler     | 369 | +1.4     | [−0.1, +1.4]   | 0.0597   | +9.0   |
|                 | all        | 369 | +1.4     | [−0.1, +1.4]   | 0.0597   | +9.0   |
| node_degree     | components | 395 | −1.0     | [−4.4, +2.6]   | 0.6543   | −1.0   |
|                 | clustering | 396 | +1.0     | [−3.0, +4.9]   | 0.7010   | +1.5   |
|                 | rwse       | 394 | +2.5     | [−1.9, +6.8]   | 0.2898   | +2.5   |
|                 | degree     | 376 | +11.7    | [+7.4, +15.0]  | 0.0001   | +8.8   |
|                 | filler     | 396 | −3.5     | [−7.3, +0.6]   | 0.0954   | −3.0   |
|                 | all        | 396 | +11.6    | [+7.3, +15.0]  | 0.0001   | +12.2  |

Table 26: qwen3-1.7b-think: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 400 | +9.2     | [+6.1, +11.3]  | 0.0001   | +9.2   |
|                 | clustering | 400 | +1.0     | [-3.1, +5.0]   | 0.6893   | +1.0   |
|                 | rwse       | 400 | +0.0     | [-4.2, +4.2]   | 1.0000   | +0.0   |
|                 | degree     | 400 | -3.0     | [-7.1, +1.4]   | 0.1882   | -3.0   |
|                 | filler     | 400 | -12.5    | [-16.2, -7.9]  | 0.0001   | -12.5  |
|                 | all        | 400 | -2.8     | [-7.0, +1.7]   | 0.2511   | -2.8   |
| cycle_check     | components | 391 | -1.3     | [-1.8, +0.3]   | 0.1318   | -2.0   |
|                 | clustering | 395 | +0.3     | [-0.2, +0.3]   | 1.0000   | +0.5   |
|                 | rwse       | 392 | +0.3     | [-0.2, +0.3]   | 1.0000   | -0.2   |
|                 | degree     | 391 | -1.5     | [-1.5, -0.1]   | 0.0312   | -2.2   |
|                 | filler     | 392 | -0.3     | [-0.3, +0.2]   | 1.0000   | -0.5   |
|                 | all        | 395 | -0.5     | [-1.0, +0.6]   | 0.6269   | -0.2   |
| edge_count      | components | 397 | -1.3     | [-2.1, +0.5]   | 0.1756   | -1.2   |
|                 | clustering | 399 | -1.3     | [-2.4, +0.6]   | 0.2298   | -1.2   |
|                 | rwse       | 398 | -1.5     | [-2.0, +0.1]   | 0.0668   | -1.5   |
|                 | degree     | 393 | +27.7    | [+24.5, +29.4] | 0.0001   | +27.3  |
|                 | filler     | 400 | +1.2     | [-0.9, +2.9]   | 0.3012   | +1.2   |
|                 | all        | 397 | +3.8     | [+1.1, +5.6]   | 0.0049   | +3.8   |
| edge_existence  | components | 400 | -2.8     | [-5.8, +0.7]   | 0.1254   | -2.8   |
|                 | clustering | 400 | +4.0     | [+0.7, +6.7]   | 0.0168   | +4.0   |
|                 | rwse       | 400 | -10.0    | [-12.9, -6.2]  | 0.0001   | -10.0  |
|                 | degree     | 398 | -4.3     | [-7.3, -0.7]   | 0.0200   | -4.8   |
|                 | filler     | 400 | -7.0     | [-10.0, -3.3]  | 0.0002   | -7.0   |
|                 | all        | 399 | -9.8     | [-12.9, -5.8]  | 0.0001   | -10.0  |
| node_count      | components | 400 | +9.0     | [+6.9, +9.5]   | 0.0001   | +9.0   |
|                 | clustering | 400 | +8.8     | [+6.2, +9.9]   | 0.0001   | +8.8   |
|                 | rwse       | 400 | +9.2     | [+6.9, +10.1]  | 0.0001   | +9.2   |
|                 | degree     | 400 | +8.0     | [+5.1, +9.7]   | 0.0001   | +8.0   |
|                 | filler     | 400 | +9.2     | [+7.5, +9.2]   | 0.0001   | +9.2   |
|                 | all        | 400 | +9.8     | [+8.0, +9.8]   | 0.0001   | +9.8   |
| node_degree     | components | 400 | +0.2     | [-0.2, +0.2]   | 1.0000   | +0.2   |
|                 | clustering | 400 | -0.5     | [-1.7, +1.0]   | 0.7246   | -0.5   |
|                 | rwse       | 400 | -2.2     | [-3.1, -0.3]   | 0.0240   | -2.2   |
|                 | degree     | 400 | -6.5     | [-7.7, -4.0]   | 0.0001   | -6.5   |
|                 | filler     | 400 | -0.5     | [-1.7, +1.0]   | 0.7232   | -0.5   |
|                 | all        | 400 | -8.0     | [-8.9, -5.6]   | 0.0001   | -8.0   |

Table 27: qwen3-4b: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 377 | +0.5     | [-0.6, +1.0]   | 0.6249   | +0.5   |
|                 | clustering | 373 | +0.0     | [-1.2, +1.2]   | 1.0000   | -0.8   |
|                 | rwse       | 379 | +0.0     | [-1.4, +1.4]   | 1.0000   | +0.5   |
|                 | degree     | 376 | +0.0     | [-1.5, +1.5]   | 1.0000   | +0.5   |
|                 | filler     | 379 | +0.8     | [-0.6, +1.3]   | 0.3787   | +1.2   |
|                 | all        | 375 | -0.5     | [-1.8, +1.1]   | 0.7306   | -1.2   |
| cycle_check     | components | 381 | +0.0     | n/a (0 disc.)  | 1.0000   | -3.2   |
|                 | clustering | 394 | +0.0     | n/a (0 disc.)  | 1.0000   | +0.0   |
|                 | rwse       | 395 | +0.0     | n/a (0 disc.)  | 1.0000   | +0.2   |
|                 | degree     | 375 | -0.8     | [-0.8, +0.3]   | 0.2514   | -5.5   |
|                 | filler     | 396 | +0.0     | n/a (0 disc.)  | 1.0000   | +0.5   |
|                 | all        | 392 | +0.0     | n/a (0 disc.)  | 1.0000   | -0.5   |
| edge_count      | components | 60  | +1.7     | [-1.6, +1.7]   | 1.0000   | +24.8  |
|                 | clustering | 59  | +1.7     | [-4.1, +5.0]   | 1.0000   | +24.8  |
|                 | rwse       | 49  | +0.0     | [-4.0, +4.0]   | 1.0000   | +10.2  |
|                 | degree     | 34  | -23.5    | [-23.5, -6.1]  | 0.0073   | +8.5   |
|                 | filler     | 50  | -2.0     | [-2.0, +1.9]   | 1.0000   | +12.2  |
|                 | all        | 41  | -41.5    | [-41.5, -25.3] | 0.0002   | +1.8   |
| edge_existence  | components | 396 | +0.0     | n/a (0 disc.)  | 1.0000   | -0.5   |
|                 | clustering | 396 | +0.0     | n/a (0 disc.)  | 1.0000   | -0.8   |
|                 | rwse       | 393 | +0.0     | n/a (0 disc.)  | 1.0000   | -1.2   |
|                 | degree     | 398 | +0.0     | n/a (0 disc.)  | 1.0000   | +0.0   |
|                 | filler     | 399 | +0.0     | n/a (0 disc.)  | 1.0000   | +0.0   |
|                 | all        | 394 | +0.0     | n/a (0 disc.)  | 1.0000   | -1.0   |
| node_count      | components | 400 | +0.0     | [-0.9, +0.9]   | 1.0000   | +0.0   |
|                 | clustering | 399 | +0.5     | [-0.3, +0.5]   | 0.4936   | +0.2   |
|                 | rwse       | 399 | +0.5     | [-0.3, +0.5]   | 0.5005   | +0.2   |
|                 | degree     | 399 | +0.5     | [-0.3, +0.5]   | 0.5005   | +0.2   |
|                 | filler     | 400 | +0.5     | [-0.3, +0.5]   | 0.4870   | +0.5   |
|                 | all        | 400 | +0.5     | [-0.3, +0.5]   | 0.4870   | +0.5   |
| node_degree     | components | 380 | -0.3     | [-0.3, +0.3]   | 1.0000   | +0.2   |
|                 | clustering | 386 | -0.3     | [-0.8, +0.6]   | 1.0000   | +1.8   |
|                 | rwse       | 381 | -2.1     | [-2.1, -0.5]   | 0.0072   | -1.2   |
|                 | degree     | 388 | +0.0     | [-0.5, +0.5]   | 1.0000   | +2.5   |
|                 | filler     | 386 | +0.0     | n/a (0 disc.)  | 1.0000   | +2.0   |
|                 | all        | 383 | -0.8     | [-0.8, +0.3]   | 0.2466   | +0.5   |

Table 28: qwen3-4b-think: all cells.